

Ezra Silver-Isenstadt

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EDUCATION

The University of Maryland, College Park | Honors College | College Park, MD

Expected May 2026

B.S. in Computer Science, B.M. in Music Composition, Minor in Philosophy | GPA: 3.9/4.0

Banneker/Key Scholar: One of ~75/57,220 awarded an all-expenses merit scholarship; featured as model B/K student for new students

Relevant Courses: Object-Oriented Programming, Computer Systems, Organization of Programming Languages, Discrete Structures, Algorithms, Web Development, Advanced Data Structures, Machine Learning, Databases, Computer and Network Security, Embodied AI

TECHNICAL SKILLS

Languages: Java, C, Python, JavaScript, TypeScript, HTML/CSS, Rust, OCaml, SQL

Tools/Frameworks: Git, React, React Native, MongoDB, Node.js, Express, Unix Command Line, Figma, JIRA

WORK EXPERIENCE

Prototype Engineer Intern | Click Therapeutics | NYC

Summer 2025

- Developed React Native prototype integrating generative AI (Q&A chatbot, text-to-speech) into a digital therapeutics platform, collaborating with Design and User Research teams and incorporating patient testing feedback to enhance therapeutic outcomes
- Built configurable breathing animation component for production component library using React Native Reanimated. Created animation sequences with custom callback functions for state transitions. Wrote Storybook documentation and Jest unit tests
- Won 1st place in an internal hackathon — awarded by the CEO — for co-developing and presenting an LLM-powered system that reduced business logic translation time by ~80%

Strategic Operations Intern | MyStage | Remote

Spring 2025

- Provide strategic guidance to the CEO on app structure, feature design, marketing approach, and other key decisions
- Conduct market research across 5+ domains and competitor analyses to inform leadership
- Collaborate with designers to align features with user needs

Lead Guitarist on Two National Tours with Lyn Lapid

Fall 2022 - Summer 2023

- Played 20 cities in the U.S. and Canada at sold-out venues like the Bowery Ballroom (NYC) and the Roxy (L.A.)
- Used organizational and communication skills to keep the production running smoothly and maintain camaraderie while circling the entire perimeter of the US in a full tour van
- Jumped in to prepare and perform a full opening set of my original music when, hours before the show in Portland, the tour's opening artist fell ill

Google Earth Engine Developer Intern | NASA Goddard Space Flight Center | Greenbelt, MD

Summer 2020

- Programmed in JavaScript using the Google Earth Engine cloud-computing platform and API
- Collaborated with mentors to create four user-friendly web applications, analyzing satellite data (including the Landsat series) to visualize climate change impacts
- Designed an intuitive UI that allows users to observe ice melt, hurricane disturbances, precipitation trends, and droughts

PROJECTS

Ike: A Chess Engine (Rust)

- Generated legal moves in parallel using bit manipulations on piece bitstrings, as well as pre-computation alongside magic bitboard hashing to find all sliding piece moves in $O(1)$, resulting in 20 million moves/sec
- Used advanced game tree search algorithms like negamax and alpha-beta pruning to find the best move

MicroCaml (OCaml)

- Developed a lexer, parser, optimizer, type-checker, and interpreter for a simple programming language
- Regex powers lexing, a context-free grammar structures parsing, subtyping and type-checking rules verify well-typed expressions, and operational semantics define interpretation
- Optimizations include constant folding, constant propagation, and short-circuiting

A Shell (C)

- Created a Unix-like shell that executes commands, manages processes, and supports input/output redirection
- Implemented control structures including pipes, subshells, and command conjunctions (AND, OR), enhancing command flexibility
- Managed file descriptors, enabling user-specified input and output files

OTHER

- Leadership:** President of Animals for Animals at UMD (2022-2024), Co-Lead of March For Our Lives, Howard County (2019-2020)
- Other Projects:** Regex acceptance with NFA to DFA (OCaml), Text editor (C), Personnel database (Java), Calendar (C)
- Other Skills:** Piano, Guitar, Music Production (Logic), Video Production (Final Cut Pro), Teaching, Public Speaking